

Advanced Review of Complex Analysis
For KAIST Graduate School Entrance Examination

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0.1 Foundations: Holomorphicity and Geometry

0.1.1 Complex Differentiability

Definition 0.1.1. Let $f: \Omega \rightarrow \mathbb{C}$ where $\Omega \subseteq \mathbb{C}$ is open.

1. f is *differentiable* at $z_0 \in \Omega$ if $\lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}$ exists.
2. f is *holomorphic* on Ω if it is differentiable at every point in Ω .
3. f is *analytic* on Ω if for every $z_0 \in \Omega$, f has a power series representation valid in some disk $B(z_0; r)$.
4. **Key Point:** In complex analysis, Holomorphic $\iff$ Analytic. (This is a deep result).

Theorem 0.1.2. Let $f(z) = u(x, y) + iv(x, y)$. f is differentiable at $z_0 = x_0 + iy_0$ if and only if u, v are differentiable in the real sense and satisfy:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Complex form: $\frac{\partial f}{\partial \bar{z}} = 0$, where $\frac{\partial}{\partial \bar{z}} = \frac{1}{2}(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y})$.

0.1.2 Harmonic Functions

Proposition 0.1.3. If $f = u + iv$ is holomorphic, then u and v are *harmonic*, i.e., $\Delta u = \nabla^2 u = 0$.

Definition 0.1.4. Given harmonic u , a harmonic function v is a *harmonic conjugate* of u if $f = u + iv$ is holomorphic.

- **Existence:** Does every harmonic u have a conjugate v on any domain? (Check: Simply connected condition).

0.2 Elementary Functions and Branch Cuts

Definition 0.2.1. $\log z = \ln |z| + i \arg z$.

1. *Principal Branch:* $\text{Log } z = \ln |z| + i\Theta$, $\Theta \in (-\pi, \pi]$.
2. *Branch Cut:* Discuss the necessity of removing a ray (e.g., negative real axis) to maintain holomorphicity.

Proposition 0.2.2. $z^c = e^{c \log z}$. Discuss how the number of values depends on whether c is an integer, a rational, or an irrational/complex number.

0.3 Complex Integration: The Heart of the Theory

0.3.1 Cauchy's Theorem

Theorem 0.3.1. If f is holomorphic on a **simply connected** domain Ω , then for any closed contour $C \subset \Omega$:

$$\oint_C f(z) dz = 0.$$

Theorem 0.3.2. If f is holomorphic in Ω , and C is a simple closed curve in Ω (positively oriented), for any z_0 inside

C :

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^{n+1}} dz.$$

0.3.2 Global Properties from Integration

Theorem 0.3.3. (Converse of Cauchy's Theorem) If $f : \Omega \rightarrow \mathbb{C}$ is continuous and $\oint_C f(z) dz = 0$ for every closed contour $C \subset \Omega$, then f is holomorphic in Ω .

Theorem 0.3.4. Every bounded entire function is constant.

- **Extension:** If $|f(z)| \leq A + B|z|^k$, then f is a polynomial of degree at most k .

0.4 Local Properties and Schwarz Lemma

0.4.1 Maximum Modulus Principle

Theorem 0.4.1. If f is holomorphic and non-constant on a domain Ω , then $|f|$ cannot attain a local maximum in Ω . If Ω is bounded and f is continuous on $\bar{\Omega}$, then $|f|$ attains its maximum on the boundary $\partial\Omega$.

0.4.2 The Schwarz Lemma (High Priority)

Lemma 0.4.2. Let $\mathbb{D} = \{z \mid |z| < 1\}$. Let $f : \mathbb{D} \rightarrow \mathbb{D}$ be holomorphic with $f(0) = 0$. Then:

1. $|f(z)| \leq |z|$ for all $z \in \mathbb{D}$.
2. $|f'(0)| \leq 1$.
3. If $|f(z)| = |z|$ for some $z \neq 0$ or $|f'(0)| = 1$, then $f(z) = e^{i\theta} z$ (a rotation).

Theorem 0.4.3. Every biholomorphic map $f : \mathbb{D} \rightarrow \mathbb{D}$ is a **Möbius transformation** of the form:

$$f(z) = e^{i\theta} \frac{z - \alpha}{1 - \bar{\alpha}z}, \quad |\alpha| < 1.$$

0.5 Singularities and Residues

0.5.1 Laurent Series and Classification

Definition 0.5.1. Let z_0 be an isolated singularity of f . Let $f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n$ be its Laurent series in $0 < |z - z_0| < R$.

1. **Removable:** $a_n = 0$ for all $n < 0$. (Limit $\lim_{z \rightarrow z_0} f(z)$ exists).
2. **Pole of order m :** $a_{-m} \neq 0$ and $a_n = 0$ for all $n < -m$. ($\lim_{z \rightarrow z_0} |f(z)| = \infty$).
3. **Essential:** Infinitely many $n < 0$ have $a_n \neq 0$. (Behavior is chaotic).

Theorem 0.5.2. If z_0 is an essential singularity, then $f(B(z_0; \epsilon) \setminus \{z_0\})$ is dense in $\mathbb{C}$.

0.5.2 Residue Calculus

Theorem 0.5.3. $\oint_C f(z) dz = 2\pi i \sum_{k=1}^n \text{Res}(f; z_k).$

Proposition 0.5.4. • Simple pole at z_0 : $\text{Res}(f; z_0) = \lim_{z \rightarrow z_0} (z - z_0)f(z).$

• Pole of order m : $\text{Res}(f; z_0) = \frac{1}{(m-1)!} \lim_{z \rightarrow z_0} \frac{d^{m-1}}{dz^{m-1}} [(z - z_0)^m f(z)].$

• If $f(z) = \frac{p(z)}{q(z)}$ and q has a simple zero at z_0 : $\text{Res}(f; z_0) = \frac{p(z_0)}{q'(z_0)}.$

0.6 Argument Principle and Rouché's Theorem

Theorem 0.6.1. Let f be meromorphic. $\frac{1}{2\pi i} \oint_C \frac{f'(z)}{f(z)} dz = Z - P$, where Z is the number of zeros and P is the number of poles inside C (counting multiplicities).

Theorem 0.6.2. If $|f(z) - g(z)| < |f(z)|$ on the contour C , then f and g have the same number of zeros inside C .

• Exam Tip: Used to locate zeros of polynomials in specific regions.

Theorem 0.6.3. If f is holomorphic and non-constant, then f maps open sets to open sets.

0.7 Conformal Mappings and Möbius Geometry

0.7.1 Conformal Properties

Definition 0.7.1. A map f is *conformal* at z_0 if it preserves angles between curves. This occurs if f is holomorphic and $f'(z_0) \neq 0$.

0.7.2 Möbius Transformations

Definition 0.7.2. $T(z) = \frac{az+b}{cz+d}$, $ad - bc \neq 0$.

1. Mapping circles/lines to circles/lines.
2. *Cross-ratio*: $(z_1, z_2, z_3, z_4) = \frac{(z_1 - z_3)(z_2 - z_4)}{(z_1 - z_4)(z_2 - z_3)}$ is invariant under T .
3. Mapping the Upper Half Plane $\mathbb{H}$ to the Unit Disk $\mathbb{D}$.

0.7.3 Riemann Mapping Theorem

Theorem 0.7.3. Given any simply connected domain $\Omega \subsetneq \mathbb{C}$ (where $\Omega \neq \mathbb{C}$), there exists a biholomorphic map $f: \Omega \rightarrow \mathbb{D}$.